

TOBIAS LIFSCHITZ

Embedded Systems Engineer

lifschitzt@gmail.com
[linkedin.com/in/tlifschitz](https://www.linkedin.com/in/tlifschitz)
github.com/tlifschitz

Senior Embedded Systems Engineer building low-power wireless sensor platforms from concept to large-scale field deployment. Deep experience in RTOS firmware, embedded DSP on constrained MCUs, and Thread mesh networks at scale. Over 6 years delivering connected hardware products, with recent ownership across mechanical, electronics, firmware, and cloud integration for industrial sensing systems.

Experience

MAPER

Buenos Aires, Argentina

Industrial IoT startup developing in-house hardware and software for predictive maintenance.

Contributed to scaling the company's wireless vibration monitoring platform from prototype to over 10,000 deployed devices.

Embedded Systems Technical Lead

Apr. 2025 – Present

- **Own technical direction** across firmware, sensor hardware, and device-cloud integration for an industrial sensor platform, mentoring a small team while staying hands-on on critical-path firmware.
- **Diagnosed and resolved** a Thread network scaling failure at our largest deployment (~800 devices): partitioned the site across 4 channels and built auto channel-scan firmware; site now runs 1,000+ devices with stable connectivity.
- **Drive next-generation sensor platform** development across mechanical, electronics, and firmware: enclosure redesign of injection-molded and machined parts for water ingress and manufacturing yield, plus a space-constrained rigid-flex PCB and a lower-power accelerometer plus magnetic field sensor. In validation. Specified requirements and ran prototype iterations with external suppliers.
- **Build field-serviceability** firmware, NFC commissioning with a mobile diagnostics app and automatic TX power tuning – targeting shorter installs, fewer unnecessary replacements, and lower power draw.

Senior Embedded Systems Engineer

Jul. 2023 – Mar. 2025

- Coded the firmware of wireless vibration sensors, sustaining 10 rollouts per year with 100% sensor adoption. This firmware has supported the scaling of the company from 0 to 6,000 sensors in the field by early 2025.
- Implemented automatic builds and test pipelines across firmware products, reducing 50% the deployment time and making them more reliable.
- Designed and implemented an advanced continuous vibration capture mechanism unlocking new business opportunities in niche markets for the company by monitoring specific asset types.

Embedded Systems Engineer

Mar. 2021 – Jun. 2023

- Performed full firmware lifecycle, from development to firmware release and OTA updates.
- Optimized vibration sensor low-power consumption, increasing battery life by 50%.
- Implemented DSP algorithms to improve the sensor monitoring features.
- Engineered an automatic testbench using a Raspberry Pi to perform end-of-line QA checks.

Embedded Systems Developer Intern

Aug. 2019 – Feb. 2021

- Defined requirements, designed hardware and firmware for a wireless vibration sensor.
- Performed PCB bring-up, and defined and executed validation tests using lab equipment.
- Developed drivers for onboard sensors and integrated OpenThread protocol stack to application code.

Education

Instituto Tecnológico de Buenos Aires (ITBA)

Buenos Aires, Argentina

Electronics Engineering

2015 – 2020

Projects: embedded linux IIOT Gateway, RISC-V core in FPGA, IMU 3D control

Awards

- **Premio Amigos del ITBA (2021)** Awarded for the second-best GPA of the cohort
- **IEEEExtreme 12.0-15.0 (2018 – 2021)** Ranked above 96% of teams worldwide (CorDamp team)
- **Electronic Fair Winner (2019)** First place for RISC-V team project in a university contest.

Skills

Embedded & Firmware – C/C++ · RTOS · ARM Cortex M4 · Embedded Linux · DMA · DSP (FFT, FIR, IIR, decimation)

Connectivity – Thread/Matter · MQTT · CoAP · UDP · Modbus · SPI · UART · I²C

Hardware & Lab – PCB design · JTAG/SWD · Oscilloscope · Power profiler

Tooling – Git · CI/CD · CMake · Make

Backend & Data – Python · Django · PostgreSQL · Grafana · Jupyter

Languages – Spanish (native) · English (fluent)